



Manifold for Type 1 Diabetes

Research-partnership brief — pre-clinical risk stratification and intervention-window detection

1 · THE CLINICAL PROBLEM

Type 1 Diabetes has a years-long pre-clinical prodrome:

- **Stage 1** — ≥ 2 persistent islet autoantibodies, normoglycemia.
- **Stage 2** — dysglycemia (abnormal OGTT), still asymptomatic.
- **Stage 3** — clinical onset (~100% 10-year risk conditional on Stage 1).

Current risk-stratification tools (TrialNet DPTRS, Index60, M120) deliver strong population-level survival curves but fail to identify **individual-trajectory inflection points** — the window within which a Stage-2-appropriate intervention (teplizumab-class anti-CD3, low-dose ATG, experimental antigen-specific therapies) delivers maximal benefit.

Three capabilities remain underdeveloped in current practice:

- Treatment of autoantibody seroconversion as a richer signal than a binary count.
- Incorporation of continuous glycemic variability *preceding* overt dysglycemia.
- Unified modeling of the causal structure between HLA haplotype, viral exposure (enterovirus / Coxsackie B), gut microbiome, and metabolic load, rather than parallel correlation studies.

Teplizumab's 2022 approval transformed the pre-clinical window from academic interest to actionable intervention. Each additional month of accurate per-patient stratification extends the period within which a Stage-2-appropriate intervention remains beneficial.

2 · WHAT MANIFOLD BRINGS

Manifold is a causal-intelligence platform for detecting structural fragility and regime transitions in complex multi-signal systems. Three engine capabilities form the basis of the T1D configuration:

1. **Causal discovery (Spirtes / PC algorithm) on multi-omic cohorts.** Replaces *ad hoc* correlation practice in GWAS × microbiome × virome × metabolomic integration with a principled conditional-independence-based directed acyclic graph.
2. **Criticality estimators feeding a composite fragility score (ΩF).** Individual estimators are configured to the domain's dynamics; the composite fragility score provides a stable integration layer across them.
3. **Pearl do-calculus and interdiction.** Therapy selection is framed as a maximum-attenuation intervention on the inferred causal graph — directly parallel to the teplizumab-class "delay onset" trial design.

3 · ΩF PILLAR MAPPING - T1D PHYSIOLOGY

ΩF PILLAR	T1D READING
I — Interaction	Autoantibody-to-autoantibody coupling (IAA, GADA, IA-2A, ZnT8); HLA × viral-trigger epistasis
R — Reserve	β -cell functional mass proxied by C-peptide AUC on MMTT; glucose-response amplitude
J — Junction	Immune–metabolic subsystem coupling (transfer entropy between immune-cell subsets and glycemic variability)
C — Cascade	Autoreactive T-cell clonal expansion; epitope spreading rate
T — Temporal	Hazard of next stage transition given current trajectory

4 · CRITICALITY ESTIMATOR CONFIGURATION

T1D pre-clinical progression is a staged Markov / survival process with autoreactive positive feedback. The criticality layer is configured accordingly:

ESTIMATOR	WHAT IT CATCHES	PILLAR FEED
Dynamic Network Biomarkers (Chen et al. 2012)	Leading-subset variance + correlation spike <i>before</i> overt stage transition	I, T
Bayesian online change-point (BOCPD)	Discrete Stage 1→2→3 transition timing with calibrated uncertainty	T, R
Deep survival with time-varying covariates (DeepSurv / RNN-Surv)	Personalized hazard given longitudinal autoantibody + CGM + metabolic signals	C, T

Additional estimators available within the same layer for subsequent phases:

- **Variance-based early-warning signals** on rolling CGM windows once continuous glucose data is onboarded — feeds the Temporal pillar.
- **Persistent homology** on longitudinal multi-omic point clouds for topological regime detection across integrated immune-metabolic trajectories — feeds the Interaction and Cascade pillars.

5 · PROPOSED INITIAL ENGAGEMENT (PHASE 1)

The scope of the proposed Phase 1 engagement is defined to produce a deliverable of demonstrable analytic value to the partnering investigator within a contained time and data footprint.

- **Dataset.** Retrospective slice of the TEDDY cohort — approximately 8,000 children, 15 years of serial autoantibody, HLA, limited metabolomic, and environmental exposure data. TrialNet PTP serves as secondary validation.
- **Access path.** NIDDK Central Repository Data Use Agreement, co-sponsored by a TEDDY-affiliated principal investigator.
- **Deliverable.** Per-subject Ω F trajectory combining DNB, change-point, and deep-survival estimators, benchmarked against DPTRS on Stage-3 prediction at 1, 3, and 5-year horizons.
- **Success criteria.** AUC lift of at least 0.05 over DPTRS on matched subjects; calibration improvement measured by Brier score and calibration-in-the-large; human-interpretable Ω F-pillar decomposition per subject.
- **Timeline.** Ten to twelve weeks from data access.
- **Excluded from Phase 1** (reserved for the Phase 2 prospective arm): live CGM ingestion, gut microbiome sequencing, virome, and fecal metabolomics.

6 · RATIONALE AND TIMING

- **Tepilizumab has been approved.** The pre-clinical window is clinically actionable; Stage-2 patients are candidates for intervention today.
- **The TEDDY cohort is mature.** Data access pathways are established, and the surrounding analytic community is sophisticated and receptive to novel methodology.
- **Relevant funding mechanisms are active.** Breakthrough T1D (formerly JDRF) Prediction & Prevention, the Helmsley Charitable Trust T1D program, NIH NIDDK SBIR/STTR, and the Breakthrough T1D Industry Discovery & Development Partnerships track all have active Requests for Proposals in this area.
- **The platform is production-ready.** Manifold's causal, criticality, and interdiction engines are production-grade; configuration for the T1D domain represents bounded, well-scoped work.

7 · PROPOSED PARTNERSHIP

Partnership with a TEDDY-affiliated principal investigator (or equivalent cohort steward — TrialNet, DiPiS, BabyDiab) is sought for:

1. Retrospective data access under the partnering institution's existing Data Use Agreement.
2. Clinical co-authorship on the benchmark manuscript.
3. Co-application to Breakthrough T1D, the Helmsley Charitable Trust, or NIH NIDDK for the Phase 2 prospective arm.

In exchange, Apex Analytica contributes the causal-inference engine, the Ω F framework, and the engineering team. Apex retains commercial rights to the platform; research intellectual property from the joint work is shared on standard academic-industry terms.